

Karnataka SIR 2026 — Draft Roll / ASD Overlap Audit

Exhaustive statewide cross-check: EPICs appearing on both the draft electoral roll and the ASD ("uncollectable elector") list

Prepared 30 August 2026 · karnataka-draft-roll-2026 project · source: `scripts/13-overlap-audit.mjs` · raw results: `test-logs/test-log.jsonl` (layer: `overlap-audit`)

1. Executive summary

This site indexes two separate public datasets published by the Election Commission of India for the Karnataka SIR 2026: the **draft electoral roll** (44,385,222 electors) and the **ASD list** (10,766,778 rows) — a report of electors whose enumeration form the Booth Level Officer (BLO) could not collect. The two are largely, but not entirely, disjoint populations.

An **exhaustive** check — every one of the two published bucket trees, not a sample — found **3,191 EPICs that genuinely appear on both lists**. This is a real, verified finding: 0.01% of the roll and 0.03% of the ASD list. It directly supersedes an earlier *sampled* check (a few thousand rows, both in this project's own prior session and in the ASD feature's original design document, `OBSERVATIONS-ASD.md`) that found zero overlaps and concluded the two lists were disjoint. That sampled conclusion was not wrong given its sample size — the true rate is low enough (about 1 in 14,000 roll electors) that a few-thousand-row sample was always unlikely to catch one. It should not be trusted again as evidence of "zero" for a rate this low.

This finding is the reason the site's search shows a dedicated "found on both lists" result, rather than assuming the two datasets can never conflict.

2. Background

The draft roll and the ASD list are generated by ECI as two separate documents and need not be perfectly synchronized in time. A voter search on this site runs both lookups independently on every query and never assumes a hit on one list rules out the other — see `docs/app.js`'s `handleEpic()` and the five-verdict design documented in the project README.

3. Methodology

Both published bucket trees (`docs/data/roll/` for the draft roll and `docs/data-asd/roll/` for the ASD list) use an identical addressing scheme — SHA-256(EPIC), a 4-hex-character shard prefix selecting the bucket file, and an 8-hex-character suffix identifying the record within it. This was verified from both datasets' own `manifest.json` files before relying on it, not assumed.

That shared scheme means the same EPIC always lands in the identically-named bucket file in both trees, which makes an exhaustive check computationally cheap:

1. List every bucket file present in *both* trees (a bucket touched by neither dataset cannot contain a collision).

2. For each shared bucket, load both files once. Records in each are already sorted by suffix (a build-time invariant of both pipelines), so the two sorted arrays are merge-compared in $O(n + m)$ — no cross-product, no hashing 10 million EPICs against a lookup structure.
3. A published bucket record deliberately carries no EPIC (by design — see `3-build-data.mjs`'s own header comment on why: publishing it would make the dataset a scrapeable EPIC table). A suffix match is therefore found purely from the published data first; only for the small set of actual collisions is `cache/rows/` and `cache/asd-rows/` then re-scanned, re-hashing each row, to recover which real EPIC produced that match.

Full source: `scripts/13-overlap-audit.mjs`. Reproducible by re-running that script against the current published trees.

4. Headline results

Metric	Value
Draft roll electors checked	44,385,222
ASD rows checked	10,766,778
Bucket files compared	65,536
EPICs found on both lists	3,191
as a share of the roll	0.01%
as a share of the ASD list	0.03%
Distinct constituencies involved	203 / 224

For scale: if the two datasets' hashes were purely random and independent, pure birthday-paradox collision noise at this volume ($44.4\text{M} \times 10.7\text{M}$ over a 48-bit address space) would predict roughly 1–2 spurious matches. Finding 3,191 — about three orders of magnitude more — confirms these are overwhelmingly genuine same-EPIC matches, not hash coincidences.

5. Breakdown by ASD reason code

Reason recorded by the BLO	Count	Share
SHIFTED — permanently shifted	2,414	75.65%
ABSENT — not traced at residence	596	18.68%
DUPLICATE — already registered elsewhere	149	4.67%
DEAD — deceased	32	1.00%

6. Same-constituency vs. cross-constituency

Pattern	Count	Share
Roll and ASD entries in the same constituency	123	3.85%
Roll and ASD entries in different constituencies	3,068	96.15%

Top 15 constituencies by collision count (roll-side AC)

Constituency	Collisions
AC 153	278
AC 152	197
AC 176	173
AC 150	148
AC 174	144
AC 151	105
AC 177	84
AC 155	83
AC 154	82
AC 175	78
AC 145	65
AC 215	54
AC 158	50
AC 181	46
AC 202	42

Collisions appear across 203 of 224 constituencies (90.6%) — broadly distributed statewide, not concentrated in one district or traceable to a single processing batch. The concentration visible above tracks roughly with constituency size (larger electorates, proportionally more collisions), not a localized data-quality issue.

7. Interpretation — two confirmed mechanisms

The two mechanisms `OBSERVATIONS-ASD.md` (the ASD feature's original design document) predicted as the only ways a real overlap could occur are both directly confirmed by this data:

7.1 Timing skew (96.1% of collisions)

The draft roll and an ASD report are two different documents, not guaranteed to be generated from the same instant-in-time snapshot. An elector who shifted, was recorded as such on their *old* constituency's ASD report, and has since been added to their *new* constituency's current roll is legitimately on both lists at once — the ASD entry is not wrong, it is simply older. This matches the 3,068 (96.15%) cross-constituency collisions almost exactly, and is the dominant mechanism (SHIFTED and ABSENT together account for 3,010 of 3,191 cases).

7.2 The DUPLICATE reason code, confirmed literally true

149 rows carry the BLO's own assertion “already registered elsewhere”. This audit confirms that assertion was correct in every one of those cases — the same EPIC really does appear on the roll elsewhere. This is the one reason code where finding a collision is not just expected but is the reason code's entire

premise.

8. The sensitive case: DEAD-reason collisions

32 EPICs were recorded by a BLO as belonging to a deceased elector on the ASD list, yet the same EPIC still appears on the current live draft roll.

This is precisely the scenario the site's "found on both lists" result was designed to handle carefully. An ASD entry is the Booth Level Officer's own recorded assertion at the time it was made — not an adjudicated fact, and not necessarily current. Possible explanations include: the roll was updated after the ASD report was frozen (e.g., an error was corrected), the BLO's record was itself mistaken, or the two documents simply describe different points in time. This audit does not attempt to arbitrate which is correct for any individual case — and the site's UI deliberately does not either. It shows both records side by side and states plainly that the two sources disagree, directing the person to their BLO or the official portal rather than asserting a verdict this project's own OCR/extraction pipeline has no authority to make.

9. Example cases

EPIC, roll location, and ASD location only — no names, in keeping with this project's existing practice of not publishing elector names outside the site's own authenticated search. All examples below are real, drawn directly from the audit's own output.

9.1 SHIFTED

EPIC	On the draft roll	On the ASD list
AAH5940382	AC120 / part 215 / serial 704	AC177 / part 180 / serial 399
XMK0335125	AC150 / part 379 / serial 985	AC155 / part 163 / serial 562
ZBG6672620	AC113 / part 64 / serial 400	AC152 / part 251 / serial 339
UZJ1227339	AC153 / part 213 / serial 456	AC151 / part 506 / serial 242

9.2 ABSENT

EPIC	On the draft roll	On the ASD list
TNH3226677	AC153 / part 434 / serial 546	AC173 / part 184 / serial 519
WEC4529582	AC176 / part 20 / serial 802	AC150 / part 335 / serial 228
FJC0908822	AC203 / part 67 / serial 542	AC201 / part 37 / serial 21
XUL5715685	AC174 / part 501 / serial 1115	AC172 / part 113 / serial 85

9.3 DUPLICATE

EPIC	On the draft roll	On the ASD list
CHW1178110	AC83 / part 217 / serial 533	AC83 / part 217 / serial 164

RAL5510557	AC42 / part 105 / serial 607	AC42 / part 161 / serial 45
RWJ1167683	AC214 / part 83 / serial 674	AC189 / part 162 / serial 22
UZD3221371	AC129 / part 67 / serial 588	AC101 / part 121 / serial 67

9.4 DEAD

EPIC	On the draft roll	On the ASD list
IWQ0004648	AC145 / part 50 / serial 505	AC113 / part 248 / serial 188
XRR7882798	AC153 / part 287 / serial 601	AC153 / part 458 / serial 4
YYV0007401	AC200 / part 172 / serial 308	AC106 / part 62 / serial 210
ZBG8031361	AC151 / part 9 / serial 448	AC152 / part 44 / serial 13

10. Implications for the site

- The five-verdict search design (draft roll only / ASD only / found on both / found on neither / coverage incomplete) is confirmed necessary by real data, not a defensive design against a hypothetical. 3,191 real people will see the "found on both, they disagree" result if they search their own EPIC.
- The site never picks a winner between the two sources for a collision, and this audit's findings support that choice — there is no single reliable signal (recency, reason code, etc.) in this data that would let an automated system safely arbitrate which source is currently correct.
- Both datasets should continue to be checked independently on every search, never short-circuiting after the first hit, exactly as implemented.

11. Recommendations / next steps

1. No pipeline or extraction change is indicated — this audit found a real property of the underlying source data, not a defect in this project's OCR, extraction, or build process.
2. Consider a live spot-check of a sample of these 3,191 EPICs through the actual site UI (not just the backend bucket data checked here) once convenient, to confirm the five-verdict UI renders the conflict case correctly end-to-end. Not done as part of this audit.
3. Re-run this audit (`scripts/13-overlap-audit.mjs`) after any future full rebuild of either dataset — the collision rate is a property of the source data at a point in time and could shift as ECI's own documents are revised.

Appendix: reproducibility

Full audit: `node scripts/13-overlap-audit.mjs`

Raw results: `test-logs/test-log.jsonl`, filter for `"layer": "overlap-audit"`

Manifests checked: `docs/data/manifest.json`, `docs/data-asd/manifest.json` (both shard depth 4, suffix length 8, verified equal before relying on the bucket-pairing shortcut)

This report: `scripts/_gen_overlap_report.py`, generated programmatically from the audit's own output — no figure in this document was hand-typed.